

Decision & Insight Intelligence Platform v3.0 — Architecture Design Document

Customer Lifecycle Prediction System

Date: 2026-08-07

Status: Design Phase — Platform Architecture

Supersedes: Decision Intelligence Service v1.0 TDD · v2.0 / v2.1 Drafts

Reference: Database Design Spec v2.1 · Prediction Service (L2 Live) · Customer State Service (L1 Live)

1. Mission Statement

The Decision & Insight Intelligence Platform is the bank's central intelligence layer. It answers *Why? What? Who? When? What next? What if?* — serving Executive Management, Marketing, Retail Banking, Relationship Managers, and Business Banking with a single, consistent source of AI-powered truth.

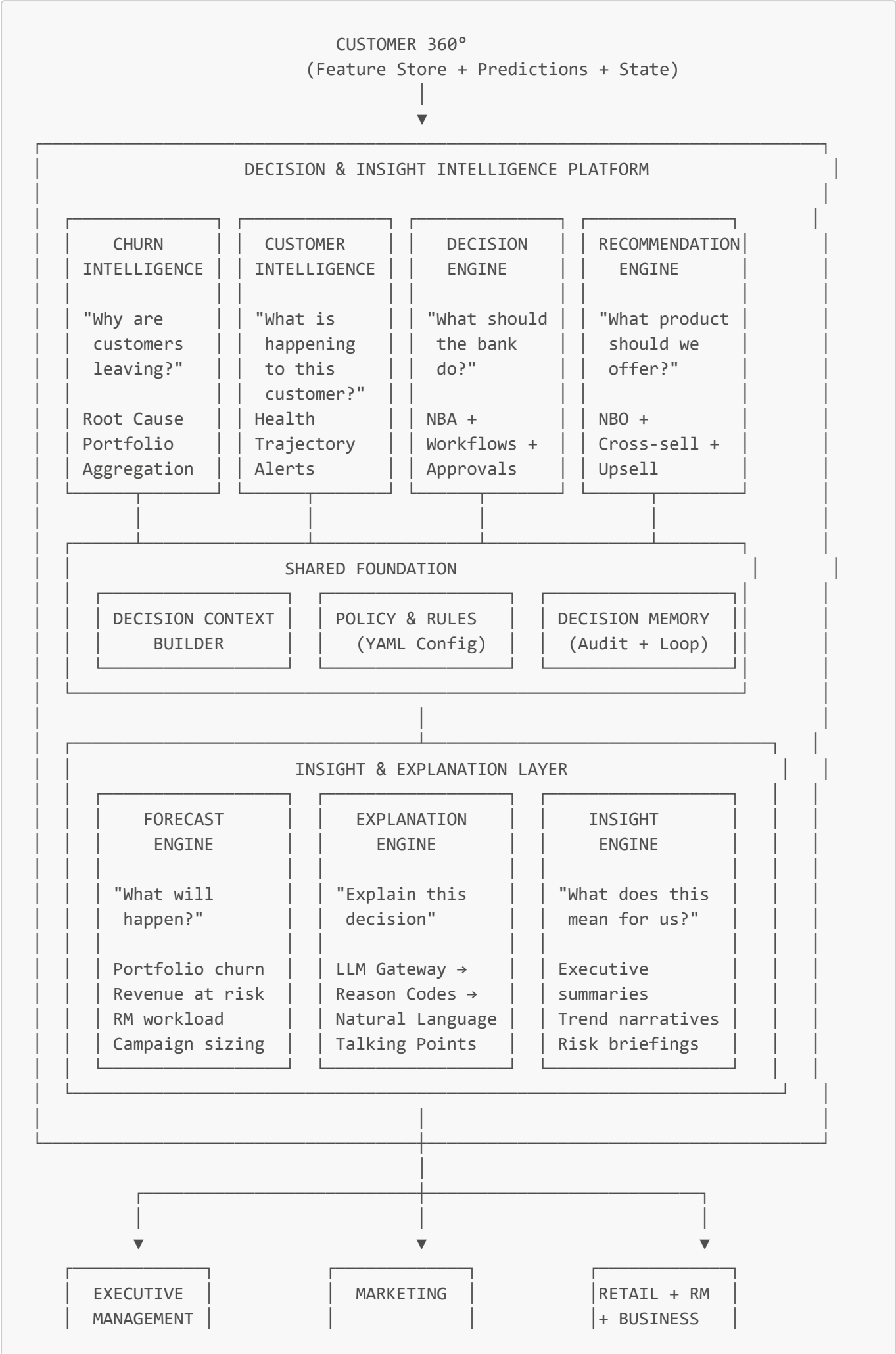
This is NOT just a Next Best Action engine. It is the platform that transforms predictions into decisions, insights, forecasts, and explanations — distributed to the right stakeholder, in the right format, at the right time.

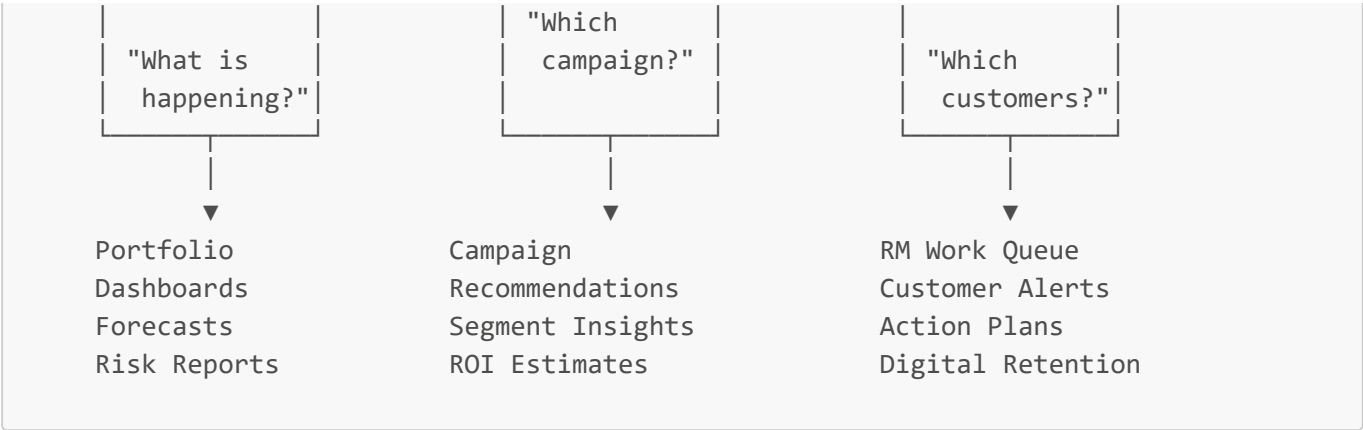
2. The Problem We're Actually Solving

Absa didn't ask for "product recommendations." They asked broader questions:

Question	Stakeholder	Engine
<i>Why are customers leaving?</i>	Executive, Marketing	Churn Intelligence
<i>Which customers are most likely to leave?</i>	Retail, RM	Customer Intelligence
<i>Which branch is losing customers?</i>	Executive, Retail	Churn Intelligence
<i>Which segment is deteriorating?</i>	Executive, Marketing	Customer Intelligence
<i>What interventions should we perform?</i>	Retail, Marketing	Decision Engine
<i>Which intervention has the highest ROI?</i>	Executive, Marketing	Recommendation + Forecast
<i>How many customers will churn next quarter?</i>	Executive	Forecast Engine
<i>Which campaigns should Marketing launch?</i>	Marketing	Recommendation Engine
<i>Which customers require RM attention?</i>	Retail, RM	Decision Engine
<i>Which customers can be retained digitally?</i>	Marketing, Retail	Decision Engine

3. Platform Architecture





4. The Six Engines

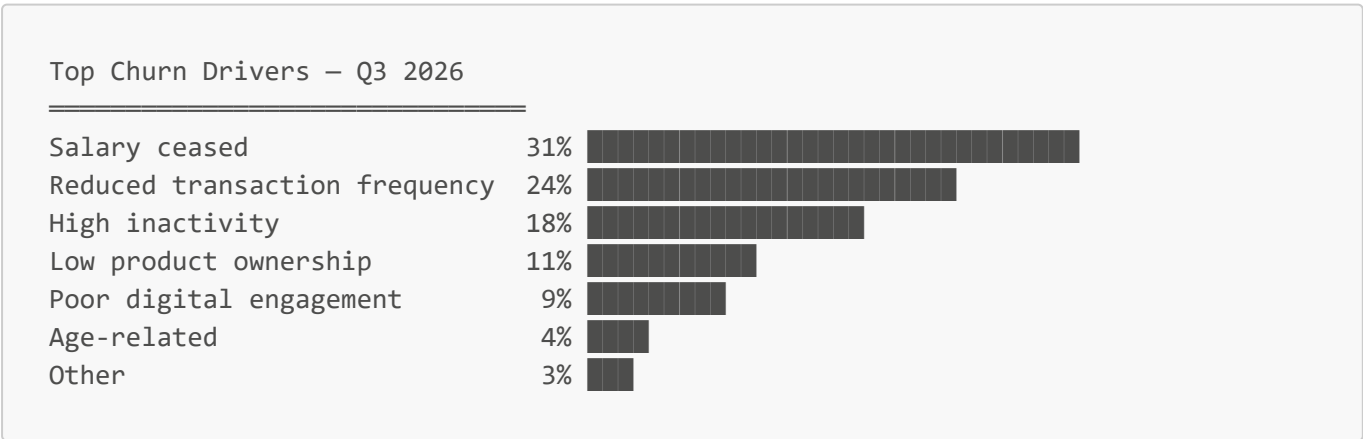
Engine 1 — Churn Intelligence

Answers: "Why are customers leaving?"

This is NOT per-customer churn probability (that's the Prediction Service). This is portfolio-level intelligence.

Capability	Description
Root Cause Analysis	Aggregate SHAP feature importance across all at-risk customers to identify top churn drivers
Segment Deterioration	Which segments are deteriorating fastest? Mass Market vs Affluent vs SME
Branch Performance	Which branches have the highest churn rates? Month-over-month trends
Driver Attribution	"Salary cessation drives 31% of churn. Reduced transaction frequency drives 24%."
Competitor Signal Detection	Patterns that suggest competitor movement (large withdrawals, salary redirects)

Example output:



LLM executive summary:

"Premier Banking churn increased 2.3pp this quarter, driven primarily by salary credit disruptions (31% of cases) and declining transaction activity (24%). The highest concentration of risk is in customers aged 30–

45 who maintain large balances but have stopped using digital channels. We recommend immediate RM outreach for the top 250 at-risk Premier customers and a targeted digital re-engagement campaign for the broader affected segment."

Engine 2 — Customer Intelligence

Answers: "What is happening to this customer?"

This is the 360° customer health narrative — not raw features, but interpreted trajectory.

Capability	Description
Health Trajectory	Is this customer improving, stable, or deteriorating? Over what timeframe?
Lifecycle Stage	Onboarding / Engaged / Mature / Declining / At Risk — with transition probabilities
Behavioural Alerts	Salary missing? Transaction volume dropping? Digital engagement declining?
Peer Comparison	How does this customer compare to their segment peers?
Predicted Trajectory	If no intervention, where will this customer be in 30/60/90 days?

Example output:

CUSTOMER INTELLIGENCE: CUST0042

Health Score: 35/100 ↓ (was 52, 30 days ago)
State: AT_RISK (transitioned from ACTIVE, 14 days ago)
Segment: Mass Affluent | Tenure: 48 months

Key Signals:

- ⚠ Salary deposit missed (last: 22 days ago, previously weekly)
- ⚠ Transaction volume ↓ 62% vs 90-day average
- ⚠ Digital logins ↓ from daily to 2x/week
- ✓ High savings balance maintained (ZMW 85,000)
- ✓ 3 active products (Savings, Current, Insurance)

90-Day Projection (no intervention):

- DORMANT: 62% probability
- CHURNED: 18% probability
- AT_RISK: 17% probability
- ACTIVE: 3% probability

Engine 3 — Decision Engine

Answers: "What should the bank do?"

This is the NBA engine from v2.1 — but now it routes decisions to the right stakeholder, not just RMs.

Capability	Description
Action Routing	Digital retention? RM call? Marketing campaign? Branch visit? Based on customer value + risk + channel preference
NBA + NBO Fusion	Combines Next Best Action (retention/service) with Next Best Offer (product)
Approval Workflow	Auto-execute low-risk actions. Require approval for high-value/FEE_WAIVER/CORPORATE
Channel Selection	Digital (SMS/email/app) vs Human (RM call/branch visit) based on customer preference + urgency
Escalation Logic	Health score <20 → Regional Manager. CLV >90th percentile + churn >0.7 → RM within 24h

Routing Matrix:

Customer Profile	Decision	Channel	Stakeholder
Premier + High CLV + High Churn	Retention Call within 48h	RM Direct	Relationship Manager
Mass Market + Moderate Churn + Mobile User	Digital Savings Campaign	SMS / App	Marketing (automated)
SME + Revenue Declining	Overdraft Review + RM Meeting	Branch Visit	Business Banking RM
Mass Affluent + Salary Missing	Salary Account Retention Offer	RM Call	Retail RM
Young Professional + Low Products + High Mobile	Digital Onboarding Campaign	App Notification	Marketing (automated)
All segments + Low Risk + Healthy	No Action / Monitor	—	System (passive)

Engine 4 — Recommendation Engine

Answers: "What product should we offer?"

This is the Next Best Offer (NBO) engine — distinct from the Decision Engine's action routing.

Capability	Description
Product Propensity	Score every product for every customer: Loan, Card, Mortgage, Insurance, Investment, FX
Cross-Sell Logic	"Customer has Savings + Salary → offer Credit Card"
Upsell Logic	"Customer has Basic Savings → offer Premium Savings"

Capability	Description
Campaign Mapping	"Recommendation: Offer Personal Loan → Campaign: PL_AUG_2026"
Eligibility-Aware	Never recommend a product the customer already holds or doesn't qualify for

Example:

PRODUCT RECOMMENDATIONS: CUST0042

Personal Loan	0.87	← Top recommendation
Credit Card	0.72	
Investment Product	0.58	
Mortgage	N/A	(customer already holds)
Insurance	N/A	(customer already holds)

Engine 5 — Forecast Engine

Answers: "What will happen?"

This is the executive-facing engine. Aggregates individual predictions into portfolio-level forecasts.

Capability	Description
Churn Forecast	Expected churn count: next month, next quarter, next 6 months
Revenue at Risk	Total CLV of customers projected to churn (ZMW)
RM Workload Forecast	How many customers will require RM intervention next week/month?
Campaign Sizing	How many customers match each campaign's target profile?
Segment Projections	Churn forecast broken down by Mass Market / Affluent / SME / Corporate
Intervention Impact	"If we execute retention calls for top 500 at-risk customers, projected churn reduction: 120 customers (24%)"

Example:

FORECAST: Q4 2026

Expected Churn:	680 customers
Revenue at Risk:	ZMW 12.4 Million
RM Workload (next 30d):	1,240 intervention candidates
Campaign Reach:	3,200 customers (Digital Retention)

By Segment:

Mass Market: 420 customers (ZMW 4.2M at risk)
Mass Affluent: 180 customers (ZMW 5.1M at risk)
Affluent: 55 customers (ZMW 2.3M at risk)
SME: 25 customers (ZMW 0.8M at risk)

Intervention Scenario:
If 500 retention calls → projected 120 saves → ZMW 2.8M preserved

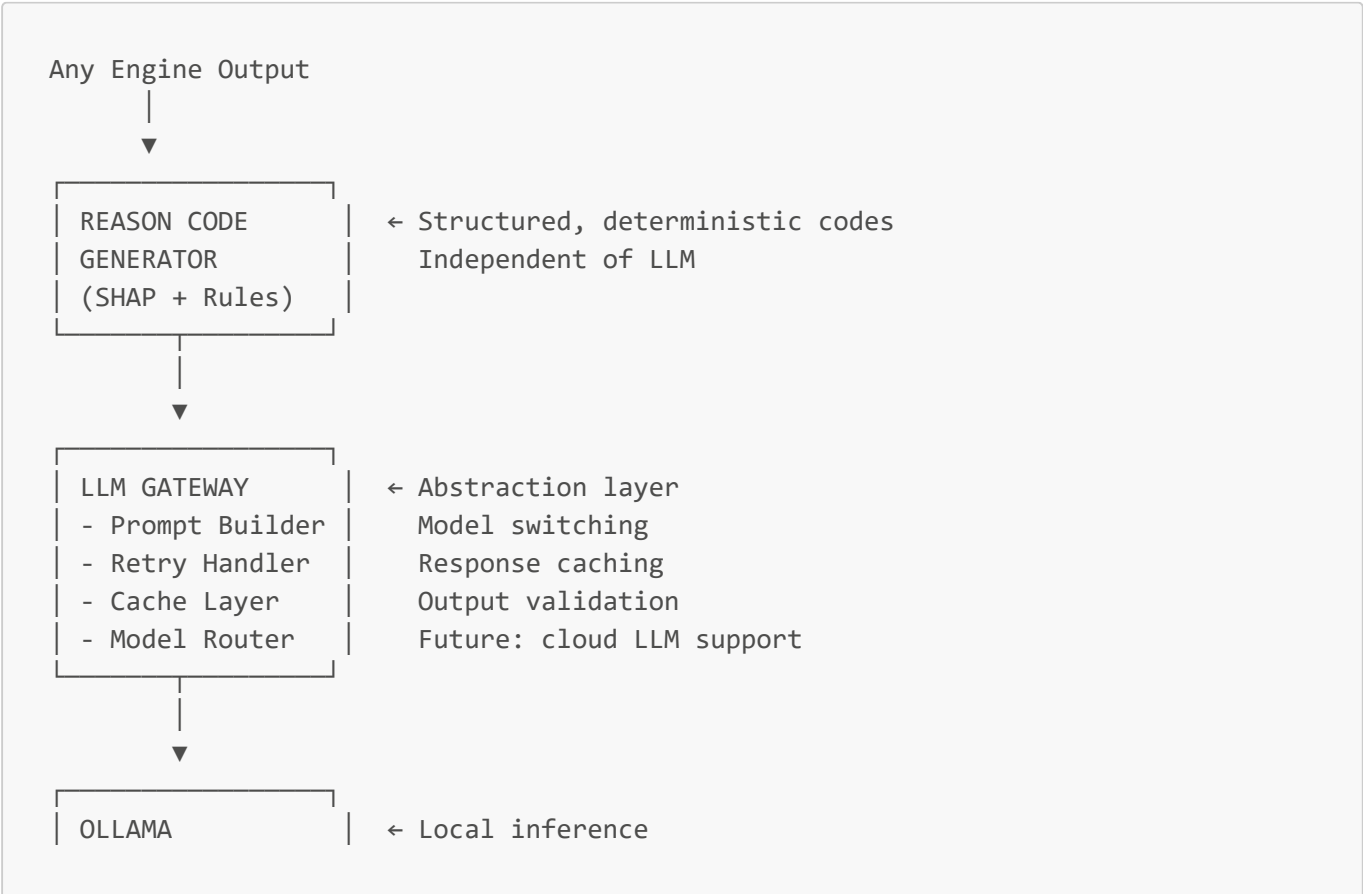
Engine 6 — Insight & Explanation Engine

Answers: "What does this mean? Explain it to me."

This is the LLM-powered layer that explains outputs from all other engines.

Capability	Description
Executive Summaries	Natural language briefings on portfolio health
Decision Explanations	Why was this NBA/NBO selected for this customer?
Trend Narratives	"Premier Banking churn increased because..."
RM Talking Points	"Here's what to say when you call this customer"
Regulatory Explanations	Full audit trail in human-readable form
Ad-Hoc Queries	"Explain why Branch 014 is underperforming"

Architecture:



(Qwen2.5 7B)

CPU-only, ~8 GB RAM

⚠ **The LLM is an ADVISOR — never a decision-maker.** It explains what the deterministic engines already produced. It never overrides business rules, ranking models, or eligibility checks.

5. Shared Foundation

All six engines share three foundation modules:

Decision Context Builder

Assembles the complete **DecisionContext** from upstream services (State, Prediction, Features). Built ONCE per customer, shared across all engines.

Policy & Rules Engine (YAML)

All business rules, eligibility checks, action catalogs, reason codes, and strategy configurations are YAML-defined and version-controlled. No hardcoded banking logic.

Decision Memory

Every decision, execution, outcome, and forecast is persisted. Powers the closed feedback loop:



6. Stakeholder Outputs

Stakeholder	Consumes From	Format	Frequency
Executive Management	Churn Intel + Forecast + Insight	Portfolio dashboards, executive briefings, risk reports	Weekly / Monthly
Marketing	Churn Intel + Recommendation + Forecast	Campaign recommendations, segment insights, ROI estimates	Weekly / On-demand
Retail Banking	Customer Intel + Decision	Customer alerts, intervention lists, health dashboards	Daily
Relationship Managers	Customer Intel + Decision + Recommendation	RM work queue, NBA/NBO, talking points, customer 360°	Daily / Real-time
Business Banking	Customer Intel + Decision + Recommendation	SME health dashboards, overdraft reviews, RM meeting prep	Weekly / On-demand

7. Internal Module Structure


```

services/decision-intelligence-platform/
├── foundation/                                # Shared by all engines
│   ├── context_builder.py                    # DecisionContext assembly
│   ├── policy_engine.py                      # YAML business rules
│   ├── eligibility_engine.py                 # YAML eligibility rules
│   └── decision_memory.py                   # Audit + feedback loop
├── engines/
│   ├── churn_intelligence/
│   │   ├── root_cause.py                    # Aggregate SHAP → portfolio drivers
│   │   ├── segment_analyzer.py              # Segment-level deterioration
│   │   └── branch_analyzer.py               # Branch-level churn metrics
│   ├── customer_intelligence/
│   │   ├── health_trajectory.py             # Health trend + trajectory
│   │   ├── lifecycle_stage.py               # Onboarding → Engaged → Declining
│   │   ├── behavioural_alerts.py            # Signal detection
│   │   └── peer_comparison.py               # Segment benchmarking
│   ├── decision_engine/
│   │   ├── action_generator.py              # 25 actions, 5 categories
│   │   ├── ranking_engine.py                # LightGBM → Top-N
│   │   ├── strategy_layer.py                # Retention First / Revenue First / etc.
│   │   ├── optimization_engine.py           # Multi-objective scoring
│   │   ├── routing_engine.py                # RM vs Digital vs Marketing
│   │   └── approval_workflow.py             # Auto vs Human approval
│   ├── recommendation_engine/
│   │   ├── product_propensity.py            # Score per product
│   │   ├── cross_sell_logic.py              # Rule-based cross-sell
│   │   ├── upsell_logic.py                  # Rule-based upsell
│   │   └── campaign_mapper.py               # NBO → Campaign
│   ├── forecast_engine/
│   │   ├── churn_forecast.py                # Aggregate → quarterly/monthly
│   │   ├── revenue_at_risk.py                # CLV × churn probability
│   │   ├── workload_forecast.py             # RM capacity planning
│   │   └── campaign_sizing.py               # Target audience sizing
│   └── insight_engine/
│       ├── reason_code_generator.py          # SHAP → structured codes
│       ├── llm_gateway.py                    # Ollama abstraction
│       ├── executive_summarizer.py           # Portfolio → narrative
│       ├── trend_narrator.py                 # Time-series → natural language
│       └── adhoc_explainer.py                # "Why is Branch 014 underperforming?"
└── api/
    ├── routes/
    │   ├── churn_intelligence.py
    │   ├── customer_intelligence.py
    │   └── decisions.py

```

```

├── recommendations.py
├── forecasts.py
├── insights.py
├── dependencies.py
├── schemas/
│   └── schemas.py # All Pydantic v2 models
├── repository/
│   └── repository.py # decisions.* schema CRUD
├── upstream/
│   └── client.py # httpx → 8002/8003/8004
├── config/
│   ├── policies/banking_rules.yaml
│   ├── eligibility/eligibility_rules.yaml
│   ├── candidates/action_catalog.yaml
│   ├── products/product_catalog.yaml
│   ├── explainability/reason_codes.yaml
│   └── strategies/strategy_config.yaml
├── prompts/
│   ├── explain_decision.txt
│   ├── executive_summary.txt
│   ├── trend_narrative.txt
│   └── adhoc_query.txt
├── tests/
│   ├── test_churn_intelligence.py
│   ├── test_customer_intelligence.py
│   ├── test_decision_engine.py
│   ├── test_recommendation_engine.py
│   ├── test_forecast_engine.py
│   └── test_pipeline.py

```

8. Decision Context (Shared Across All Engines)

```

class DecisionContext(BaseModel):
    """The single object built once and shared across all 6 engines."""

    # === Identity ===
    customer_id: str
    as_of_date: date

    # === From State Service (L1) ===
    customer_state: Literal["ACTIVE", "AT_RISK", "DORMANT", "CHURNED"]
    state_duration_days: int
    state_probability: float
    previous_state: str | None = None

```

```
# === From Prediction Service (L2) ===
health_score: float
churn_probability: float
clv_percentile: float
component_scores: dict

# === From Feature Store ===
segment: str
age: int
income_band: str | None = None
tenure_months: int
branch_code: str
preferred_channel: str | None = None
engagement_score: float | None = None
days_since_last_txn: int | None = None
total_amount_90d: float | None = None
products: dict
shap_values: dict | None = None

# === Relationship ===
rm_id: str | None = None

# === Eligibility ===
aml_flag: bool = False
kyc_expired: bool = False
marketing_opt_out: bool = False
credit_risk_rating: str | None = None
loan_in_arrears: bool = False

# === History ===
last_action_date: date | None = None
contact_frequency_30d: int = 0
previous_offers: list[dict] = []
```

9. API Contract

Churn Intelligence

Method	Endpoint	Purpose
GET	/churn-intel/drivers? as_of_date=&segment=&branch=	Top churn drivers with SHAP aggregation
GET	/churn-intel/segments?as_of_date=	Segment deterioration rankings
GET	/churn-intel/branches?as_of_date=	Branch churn performance
GET	/churn-intel/summary?as_of_date=	LLM executive summary

Customer Intelligence

Method	Endpoint	Purpose
GET	/customer-intel/{id}?as_of_date=	Full 360° customer intelligence
GET	/customer-intel/{id}/trajectory	90-day projected trajectory
GET	/customer-intel/{id}/alerts	Active behavioural alerts
GET	/customer-intel/{id}/peers	Segment peer comparison

Decision Engine

Method	Endpoint	Purpose
POST	/decisions/compute?as_of_date=&strategy=BALANCED	Batch decision computation
GET	/decisions/{id}?as_of_date=	Full decision package
GET	/decisions/queue?rm_id=&limit=20	RM priority work queue
POST	/decisions/{id}/execute	Log RM action
PATCH	/decisions/{id}/outcome	Record outcome
PATCH	/decisions/{id}/approve	Human approval

Recommendation Engine

Method	Endpoint	Purpose
GET	/recommendations/{id}?as_of_date=	Product recommendations for customer
GET	/recommendations/campaigns?segment=&limit=1000	Campaign target lists

Forecast Engine

Method	Endpoint	Purpose
GET	/forecasts/churn?horizon=90d&segment=	Churn forecast
GET	/forecasts/revenue-at-risk?horizon=90d	Revenue at risk (ZMW)
GET	/forecasts/workload?horizon=30d	RM workload projection
GET	/forecasts/campaign-sizing?campaign=	Campaign audience sizing
GET	/forecasts/scenario?intervention=retention_calls&coverage=500	What-if scenario

Insight Engine

Method	Endpoint	Purpose
GET	/insights/executive-summary?period=Q3	LLM executive briefing
GET	/insights/trend? metric=churn&segment=Premier	Natural language trend narrative
POST	/insights/explain	Ad-hoc query: "Why is Branch 014 underperforming?"

Platform

Method	Endpoint	Purpose
GET	/strategies	List active strategies
PATCH	/strategies/activate	Switch strategy
GET	/health	Platform health + upstream status
POST	/rules/reload	Reload YAML configs

10. Database Schema — **decisions** Schema

```
-- Churn Intelligence
CREATE TABLE decisions.churn_drivers (
  id BIGSERIAL PRIMARY KEY,
  as_of_date DATE NOT NULL,
  segment VARCHAR(32),
  branch_code VARCHAR(16),
  driver_rank INT,
  driver_name VARCHAR(128),
  driver_contribution_pct NUMERIC(5,2),
  affected_customer_count INT,
  computed_at TIMESTAMPTZ DEFAULT NOW()
);

-- Customer Intelligence
CREATE TABLE decisions.customer_intelligence (
  id BIGSERIAL PRIMARY KEY,
  customer_id VARCHAR(64) NOT NULL,
  as_of_date DATE NOT NULL,
  health_trajectory VARCHAR(16), -- IMPROVING, STABLE, DECLINING, CRITICAL
  lifecycle_stage VARCHAR(32),
  active_alerts JSONB DEFAULT '[]',
  projected_state_30d VARCHAR(16),
  projected_state_90d VARCHAR(16),
  peer_percentile_health NUMERIC(5,2),
  peer_percentile_engagement NUMERIC(5,2),
  computed_at TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE (customer_id, as_of_date)
);
```

```

-- Decisions (NBA)
CREATE TABLE decisions.recommendation (
  decision_id VARCHAR(64) PRIMARY KEY,
  customer_id VARCHAR(64) NOT NULL,
  as_of_date DATE NOT NULL,
  status VARCHAR(32) DEFAULT 'GENERATED',
  top_actions JSONB NOT NULL DEFAULT '[]',
  routing VARCHAR(32), -- RM, DIGITAL, MARKETING, BRANCH
  priority_score NUMERIC(5,2),
  estimated_revenue_zmw NUMERIC(12,2),
  reason_codes JSONB DEFAULT '[]',
  explanation JSONB DEFAULT '{}',
  decision_confidence JSONB DEFAULT '{}',
  approval_required BOOLEAN DEFAULT false,
  strategy VARCHAR(32) DEFAULT 'BALANCED',
  computed_at TIMESTAMPTZ DEFAULT NOW()
);

-- Recommendations (NBO)
CREATE TABLE decisions.product_recommendation (
  id BIGSERIAL PRIMARY KEY,
  customer_id VARCHAR(64) NOT NULL,
  as_of_date DATE NOT NULL,
  product VARCHAR(64) NOT NULL,
  propensity_score NUMERIC(5,4),
  is_eligible BOOLEAN,
  campaign VARCHAR(64),
  computed_at TIMESTAMPTZ DEFAULT NOW()
);

-- Forecasts
CREATE TABLE decisions.forecast (
  id BIGSERIAL PRIMARY KEY,
  forecast_date DATE NOT NULL,
  horizon_days INT NOT NULL,
  metric VARCHAR(64), -- CHURN, REVENUE_AT_RISK, WORKLOAD, CAMPAIGN_SIZE
  segment VARCHAR(32),
  predicted_value NUMERIC(15,2),
  confidence_lower NUMERIC(15,2),
  confidence_upper NUMERIC(15,2),
  computed_at TIMESTAMPTZ DEFAULT NOW()
);

-- Execution & Feedback (shared)
CREATE TABLE decisions.recommendation_execution (
  id BIGSERIAL PRIMARY KEY,
  decision_id VARCHAR(64) REFERENCES decisions.recommendation(decision_id),
  customer_id VARCHAR(64) NOT NULL,
  rm_id VARCHAR(64),
  action_taken VARCHAR(64),
  channel_used VARCHAR(32),
  status VARCHAR(32) DEFAULT 'PENDING',
  notes TEXT,
  executed_at TIMESTAMPTZ,

```

```
        created_at TIMESTAMPTZ DEFAULT NOW()
    );

    CREATE TABLE decisions.recommendation_feedback (
        id BIGSERIAL PRIMARY KEY,
        decision_id VARCHAR(64) REFERENCES decisions.recommendation(decision_id),
        customer_id VARCHAR(64) NOT NULL,
        outcome VARCHAR(32) NOT NULL,
        actual_revenue_zmw NUMERIC(12,2),
        product_adopted VARCHAR(64),
        retained_after_30d BOOLEAN,
        retained_after_90d BOOLEAN,
        feedback_date DATE,
        created_at TIMESTAMPTZ DEFAULT NOW()
    );
```

11. Phase 1 Implementation Plan

Phase 1 delivers:

Engine	Capabilities
Foundation	Decision Context Builder, Policy Engine, Eligibility Engine, Decision Memory
Decision Engine	Action Generation, Heuristic Ranking, Strategy Layer, Routing, Approval Workflow
Customer Intelligence	Health Trajectory, Lifecycle Stage, Behavioural Alerts
Recommendation Engine	Product Propensity (heuristic), Cross-sell Logic, Campaign Mapping
Churn Intelligence	Root Cause Analysis (aggregate SHAP), Segment Deterioration
Forecast Engine	Churn Forecast, Revenue at Risk
Insight Engine	Reason Code Generator (LLM Gateway + Explanation deferred to Phase 3)

Phase 1 defers:

- LightGBM ranking model (Phase 2 — heuristic scoring for now)
- LLM Gateway + Ollama explanations (Phase 3)
- Peer comparison (Phase 3)
- What-if scenario simulation (Phase 4)
- Feedback loop back to Feature Store (Phase 4)
- Campaign integration (Phase 4)

12. Technology Stack

Component	Technology	Status
Language	Python 3.12	 In .venv

Component	Technology	Status
API	FastAPI	✓ Installed
ML	LightGBM, XGBoost, SHAP	✓ Installed
Rule Engine	Custom YAML-based	✗ To build
LLM Runtime	Ollama (Qwen2.5 7B)	✗ To configure
Database	PostgreSQL (etl_clean)	✓ Running
Cache	Redis	✓ Installed
HTTP Client	httpx (upstream + Ollama)	✓ Installed
Config	PyYAML	✗ To install

13. Architecture Decision Record

#	Decision	Rationale
ADR-001	Six engines, one platform, shared foundation	Avoids code duplication. Decision Context built once, consumed by all engines
ADR-002	Churn Intelligence is portfolio-level, not per-customer	Per-customer churn is Prediction Service (L2). This engine answers "why" at scale
ADR-003	Decision routing by stakeholder, not just RM	Not every decision needs human intervention. Route to Digital/Marketing/RM based on value + urgency
ADR-004	Forecast Engine separate from individual predictions	Aggregation logic is different from per-customer scoring. Enables scenario modeling
ADR-005	LLM explains engine outputs, never makes decisions	Deterministic engines produce the decision. LLM converts to natural language
ADR-006	YAML for all rules, catalogs, strategies, codes	Non-technical stakeholders can review. Audit-friendly. Hot-reloadable
ADR-007	Decision Memory for closed-loop learning	Without feedback, models never improve. Every decision → execution → outcome tracked
ADR-008	Single DecisionContext shared across engines	No engine queries upstream services individually. Testable with mocked context

14. Future Phases

Phase	Capabilities
Phase 1	Foundation + Decision Engine + Customer Intel + Recommendation (heuristic) + Churn Intel (aggregation) + Forecast (basic)
Phase 2	LightGBM Ranking + Optimization Engine + Simulation Engine + Advanced Forecast (confidence bands)

Phase	Capabilities
Phase 3	LLM Gateway + Reason Codes + Natural Language Explanations + Executive Summaries + Ad-Hoc Queries
Phase 4	Feedback Loop → Feature Store + Campaign Integration + Full Approval Workflow + Peer Comparison
v4.0	Reinforcement Learning ranking + Real-time streaming (Kafka) + Multi-arm bandit optimization + Autonomous decisions with human-in-the-loop